

Railway Reservation System

Test Plan Document

UC1: User Registration & Login

Covers: FR1, FR2

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	UC1: User Registration & Login	Validate username field accepts only permitted characters (5-20 alphanumeric)	On registration page	1. Enter username with special characters 2. Submit	Username = "Ram@123"	Error "Invalid username format" displayed	As expected	Pass
UT_02	UC1: User Registration & Login	Validate password field enforces minimum length/complexity	On registration page	1. Enter a weak password 2. Submit	Password = "1234"	Error "Password must be at least 8 characters"	As expected	Pass
UT_03	UC1: User Registration & Login	Validate mandatory field check on empty username at login	On login page	1. Leave username blank 2. Click Login	Username = ""	Error "Username is required"	As expected	Pass
IT_01	UC1: User Registration & Login	Verify login authenticates user against credentials stored in DB (FR2)	Registered user exists	1. Enter valid username & password 2. Click Login	Username="PESU_Student01", Password="Pes@1234"	Credentials validated against DB; user logged in	As expected	Pass
IT_02	UC1: User Registration & Login	Verify logout terminates the active session (FR2)	User logged in	1. Click Logout 2. Attempt to access account page	N/A	Session ended; redirected to login page	As expected	Pass
IT_03	UC1: User Registration & Login	Verify registration creates a new account record in the database (FR1)	Unique email, valid details	1. Fill registration form 2. Submit	Name="Aditi Rao", Email="aditi.rao@test.com"	New account record created; confirmation shown	As expected	Pass
ST_01	UC1: User Registration & Login	End-to-end successful login with valid credentials (FR2)	Registered account exists	1. Navigate to login page 2. Enter valid credentials 3. Submit	Username="PESU_Student01", Password="Pes@1234"	User redirected to home/dashboard	As expected	Pass
ST_02	UC1: User Registration & Login	End-to-end failed login with invalid credentials	On login page	1. Enter valid username, wrong password 2. Submit	Password = "wrongpass"	Error "Invalid username or password"	As expected	Pass
ST_03	UC1: User Registration & Login	End-to-end new account registration followed by first login (FR1, FR2)	New email, not registered	1. Register with new details 2. Log in with the same credentials	Email = "newuser01@test.com"	Registration succeeds; login succeeds with same credentials	As expected	Pass

UC2: Search Trains & Display Results

Covers: FR3, FR4, FR5, FR6, FR7

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_04	UC2: Search Trains & Display Results	Validate source and destination cannot be the same station (FR3)	On search page	1. Select same station in Source & Destination 2. Click Search	Source = Destination = "Bangalore"	Error "Source and destination cannot be the same"	As expected	Pass
UT_05	UC2: Search Trains & Display Results	Validate journey date field rejects past dates (FR3)	On search page	1. Select a date earlier than today 2. Click Search	Date = "01-01-2020"	Error "Please select a valid future date"	As expected	Pass
UT_06	UC2: Search Trains & Display Results	Validate empty source field triggers validation error (FR3)	On search page	1. Leave Source blank 2. Click Search	Source = ""	Error "Source station is required"	As expected	Pass
IT_04	UC2: Search Trains & Display Results	Verify search returns trains matching source, destination & date from DB (FR3, FR4)	Schedule data exists in DB	1. Enter valid source/destination/date 2. Click Search	Source="Bangalore", Destination="Chennai", Date="15-09-2026"	Matching trains fetched from DB and displayed	As expected	Pass
IT_05	UC2: Search Trains & Display Results	Verify search results display train number, name, timings and fare correctly (FR5)	Matching trains exist	1. Perform a valid search 2. Inspect result fields	Train = "12658"	Train no., name, source, destination, timings & fare all shown correctly	As expected	Pass
IT_06	UC2: Search Trains & Display Results	Verify seat availability shown per class is fetched in real time (FR6)	DB seat table populated	1. Search a train 2. Check seat availability per class	Train = "12658"	Live seat counts per class match DB values	As expected	Pass
ST_04	UC2: Search Trains & Display Results	End-to-end search for a valid route and date returns matching trains (FR4)	Route exists	1. Enter valid source, destination, date 2. Click Search	Source="Bangalore", Destination="Mumbai", Date="18-09-2026"	List of matching trains with details shown	As expected	Pass
ST_05	UC2: Search Trains & Display Results	End-to-end search for a route with no trains shows "No trains found" (FR4)	No route exists between stations	1. Search for unconnected stations	Source="Bangalore", Destination="Leh"	Message "No trains found for selected route"	As expected	Pass
ST_06	UC2: Search Trains & Display Results	End-to-end selection of a train and travel class from search results (FR7)	Search results displayed	1. Select a train 2. Select a travel class	Train="12658", Class="3A"	Selected train and class carried forward to next step	As expected	Pass

UC3: Passenger Details & Fare Calculation

Covers: FR8, FR9

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_07	UC3: Passenger Details & Fare Calculation	Validate passenger age field accepts only positive integers (FR8)	On passenger details form	1. Enter negative age 2. Submit	Age = -5	Error "Enter a valid age"	As expected	Pass
UT_08	UC3: Passenger Details & Fare Calculation	Validate mandatory passenger name field (FR8)	On passenger details form	1. Leave passenger name blank 2. Click Proceed	Name = ""	Error "Passenger name is required"	As expected	Pass
UT_09	UC3: Passenger Details & Fare Calculation	Validate fare calculation formula for a given distance/class (FR9)	Fare rule loaded	1. Call fare calculation function	Distance=350km, Class="3A"	Correct fare per fare rule returned	As expected	Pass
IT_07	UC3: Passenger Details & Fare Calculation	Verify fare is recalculated correctly when travel class is changed (FR9)	Passenger details entered	1. Change class from SL to 3A 2. Check fare	Class changed SL - > 3A	Fare updates to reflect 3A pricing	As expected	Pass
IT_08	UC3: Passenger Details & Fare Calculation	Verify passenger details are validated before proceeding to payment (FR8)	Passenger form partially filled	1. Leave a mandatory field blank 2. Click Proceed to Payment	Age field blank	User blocked from proceeding; validation error shown	As expected	Pass
IT_09	UC3: Passenger Details & Fare Calculation	Verify total fare accounts for number of passengers entered (FR9)	Multiple passengers added	1. Add 3 passengers 2. Check total fare	3 passengers, Class="SL"	Total fare = per-seat fare x number of passengers	As expected	Pass
ST_07	UC3: Passenger Details & Fare Calculation	End-to-end entry of valid passenger details with computed fare displayed (FR8, FR9)	Train and class selected	1. Enter valid passenger details 2. View computed fare	Passenger="Rahul Sharma", Age=28	Fare displayed matches class and passenger count	As expected	Pass
ST_08	UC3: Passenger Details & Fare Calculation	End-to-end attempt to proceed with invalid/incomplete passenger details (FR8)	On passenger details form	1. Enter invalid age (0) 2. Click Proceed	Age = 0	Booking blocked with validation error	As expected	Pass
ST_09	UC3: Passenger Details & Fare Calculation	End-to-end fare display verified across multiple passengers/classes (FR9)	Multiple passengers, different classes not allowed per booking	1. Enter 2 passengers in same class 2. Verify fare breakdown	2 passengers, Class="2A"	Fare breakdown per passenger and total shown correctly	As expected	Pass

UC4: Payment Processing

Covers: Actor: Payment System (flow diagram)

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_10	UC4: Payment Processing	Validate payment amount matches the calculated ticket fare exactly	Fare calculated	1. Initiate payment 2. Compare requested amount to fare	Ticket fare = ₹450	Payment request amount = ₹450 exactly	As expected	Pass
UT_11	UC4: Payment Processing	Validate card/UPI input field format	On payment page	1. Enter invalid UPI ID format	UPI ID = "invalid_id"	Error "Enter a valid UPI ID"	As expected	Pass
UT_12	UC4: Payment Processing	Validate payment retry option triggers correctly on failure	Payment attempt failed	1. Simulate a failed payment 2. Check retry option	Simulated decline	"Retry Payment" option presented to user	As expected	Pass
IT_10	UC4: Payment Processing	Verify a successful payment response triggers the ticket generation step	Payment submitted	1. Complete payment via gateway 2. Check next step triggered	Amount = ₹450	Flow proceeds to "Generate Ticket" step	As expected	Pass
IT_11	UC4: Payment Processing	Verify a failed payment returns the user to the payment step without confirming booking	Payment submitted	1. Simulate payment failure 2. Check booking status	Card declined response	Booking remains unconfirmed; user returned to payment step	As expected	Pass
IT_12	UC4: Payment Processing	Verify payment status is recorded correctly against the booking attempt	Payment attempted	1. Complete or fail a payment 2. Check payment status record	Booking attempt ID = 2001	Payment status (Success/Failed) recorded against booking attempt	As expected	Pass
ST_10	UC4: Payment Processing	End-to-end successful payment via UPI/card completing a booking	Passenger details entered	1. Proceed to payment 2. Enter valid payment details 3. Submit	Amount = ₹450, valid test card	Payment successful; flow proceeds to ticket generation	As expected	Pass
ST_11	UC4: Payment Processing	End-to-end failed payment (declined) with booking not confirmed	Passenger details entered	1. Proceed to payment 2. Use a declined test card	Test card = decline scenario	Error shown; booking not confirmed	As expected	Pass
ST_12	UC4: Payment Processing	End-to-end retry payment after an initial failure succeeds	Payment previously failed	1. Retry payment with valid details 2. Submit	Retry with valid test card	Second attempt succeeds; booking proceeds to confirmation	As expected	Pass

UC5: Book Ticket & Generate PNR

Covers: FR10, FR11, FR12

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_13	UC5: Book Ticket & Generate PNR	Validate booking is blocked when no seats are available (FR10)	Coach fully booked	1. Attempt to book on a fully booked coach	Available seats = 0	Booking blocked; error "No seats available"	As expected	Pass
UT_14	UC5: Book Ticket & Generate PNR	Validate PNR is generated as a unique identifier (FR11)	Booking function called	1. Call PNR generation function	N/A	Unique PNR value returned	As expected	Pass
UT_15	UC5: Book Ticket & Generate PNR	Validate PNR is not generated for an incomplete/unpaid booking (FR11)	Payment not completed	1. Trigger PNR generation before payment success	Payment status = "Pending"	PNR generation blocked; error returned	As expected	Pass
IT_13	UC5: Book Ticket & Generate PNR	Verify booking updates seat availability in the database (FR10)	Seats available	1. Book 1 seat 2. Check DB seat count	Train="12658", Class="SL"	Available seats reduced by 1 in DB	As expected	Pass
IT_14	UC5: Book Ticket & Generate PNR	Verify the generated PNR is linked correctly to the booking record (FR11)	Booking completed	1. Complete a booking 2. Check DB record	Booking ID = 1042	PNR field populated correctly in Bookings table	As expected	Pass
IT_15	UC5: Book Ticket & Generate PNR	Verify booking confirmation displays correct ticket details after generation (FR12)	Booking confirmed	1. Complete booking 2. View confirmation screen	Booking ID = 1042	Confirmation shows PNR, passenger, train, seat & fare details	As expected	Pass
ST_13	UC5: Book Ticket & Generate PNR	End-to-end successful ticket booking with valid data and PNR generated (FR10, FR11, FR12)	Logged in, seats available, payment successful	1. Search train 2. Enter passenger details 3. Pay 4. Confirm booking	Passenger="Rahul Sharma", Class="SL"	Booking confirmed; unique PNR and ticket details displayed	As expected	Pass
ST_14	UC5: Book Ticket & Generate PNR	End-to-end booking attempt when seats are unavailable is rejected (FR10)	Train fully booked	1. Attempt booking on fully booked train	Train = "12659"	Booking rejected with "No seats available" message	As expected	Pass
ST_15	UC5: Book Ticket & Generate PNR	End-to-end view of the booking confirmation screen with all ticket details (FR12)	Booking just completed	1. Complete booking 2. Review confirmation screen	Booking ID = 1042	All required ticket details (PNR, train, seat, fare) visible	As expected	Pass

UC6: View Booking History

Covers: FR13

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_16	UC6: View Booking History	Validate booking list distinguishes upcoming from previous bookings (FR13)	Multiple bookings with different dates	1. Call history fetch function 2. Check grouping	Bookings with past & future dates	Bookings correctly grouped as "Previous" / "Upcoming"	As expected	Pass
UT_17	UC6: View Booking History	Validate empty-state message for a user with no bookings (FR13)	New user, no bookings	1. Open booking history page	User = new, 0 bookings	Message "No bookings found" displayed	As expected	Pass
UT_18	UC6: View Booking History	Validate booking entries are restricted to the logged-in user	Two users each with bookings	1. Fetch history with User A's session	User A ID	Only User A's bookings returned	As expected	Pass
IT_16	UC6: View Booking History	Verify booking history integrates with the bookings database table (FR13)	Bookings exist in DB	1. Open booking history 2. Compare with DB records	User ID = 501	All DB records for user displayed correctly	As expected	Pass
IT_17	UC6: View Booking History	Verify booking history reflects status changes such as cancellation (FR13, FR17)	Booking status changes	1. Cancel a booking 2. Refresh booking history	PNR = "1234567890"	Status updates to "Cancelled" in history list	As expected	Pass
IT_18	UC6: View Booking History	Verify booking history correctly separates previous and upcoming bookings (FR13)	Bookings on both past & future dates exist	1. Open booking history 2. Check section split	User ID = 501	Bookings displayed under correct section by journey date	As expected	Pass
ST_16	UC6: View Booking History	End-to-end view of previous and upcoming bookings for a user (FR13)	User has made 3+ bookings	1. Log in 2. Navigate to Booking History	User ID = 501	All bookings listed, correctly split into previous/upcoming	As expected	Pass
ST_17	UC6: View Booking History	End-to-end view of booking history containing mixed statuses	User has confirmed and cancelled bookings	1. Open booking history 2. Check status column	User ID = 501	Each entry shows correct status (Confirmed/Cancelled)	As expected	Pass
ST_18	UC6: View Booking History	End-to-end attempt to access another user's booking history is denied	Logged in as User A	1. Attempt to fetch User B's history via URL/ID tampering	User A tries User B's ID = 502	Access denied; "Unauthorized" error shown	As expected	Pass

UC7: Cancel Ticket & Refund

Covers: FR14, FR15, FR16, FR17

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_19	UC7: Cancel Ticket & Refund	Validate cancellation confirmation prompt is mandatory before cancelling (FR14)	On cancellation page	1. Attempt to cancel without confirming 2. Check behavior	No confirmation given	Cancellation not processed until confirmed	As expected	Pass
UT_20	UC7: Cancel Ticket & Refund	Validate refund amount calculation logic (FR16)	Cancellation function called	1. Call refund calculation function	Fare = ₹450	Refund amount returned per system-defined cancellation policy	As expected	Pass
UT_21	UC7: Cancel Ticket & Refund	Validate cancellation is blocked for a non-existent booking	On cancellation page	1. Attempt to cancel using an invalid/unknown PNR	PNR = "0000000000"	Error "Booking not found"	As expected	Pass
IT_19	UC7: Cancel Ticket & Refund	Verify cancellation updates seat availability back to the pool (FR15)	Confirmed ticket exists	1. Cancel a confirmed ticket 2. Check seat availability	PNR = "1234567890"	Seat count incremented by 1 in DB	As expected	Pass
IT_20	UC7: Cancel Ticket & Refund	Verify cancellation updates booking status to "Cancelled" in DB (FR14, FR17)	Confirmed ticket exists	1. Cancel ticket 2. Query booking record	PNR = "1234567890"	Booking status updated to "Cancelled" in DB	As expected	Pass
IT_21	UC7: Cancel Ticket & Refund	Verify cancellation validates that the booking belongs to the logged-in user	Booking belongs to a different user	1. Attempt to cancel another user's PNR	PNR belonging to User B	Error "This booking does not belong to your account"	As expected	Pass
ST_19	UC7: Cancel Ticket & Refund	End-to-end cancellation of a confirmed ticket with refund amount displayed (FR14, FR16, FR17)	Confirmed ticket exists	1. Open booking history 2. Cancel ticket 3. Confirm	PNR = "1234567890"	Ticket cancelled; refund amount and status displayed	As expected	Pass
ST_20	UC7: Cancel Ticket & Refund	End-to-end attempt to cancel a ticket that does not belong to the user	Logged in as User A	1. Attempt to cancel User B's ticket	PNR belonging to User B	Cancellation denied; error shown	As expected	Pass
ST_21	UC7: Cancel Ticket & Refund	End-to-end attempt to cancel an already-cancelled ticket	Ticket already cancelled	1. Attempt to cancel the same PNR again	PNR = "1234567890"	Error "Ticket already cancelled"	As expected	Pass

UC8: Admin – Manage Trains, Routes & Schedules

Covers: FR18, FR19, FR20, FR21

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_22	UC8: Admin – Manage Trains, Routes & Schedules	Validate train number field uniqueness on add (FR18)	Admin on Add Train form	1. Enter an already-existing train number 2. Submit	Train No. = "12658" (existing)	Error "Train number already exists"	As expected	Pass
UT_23	UC8: Admin – Manage Trains, Routes & Schedules	Validate mandatory fields on the Add Train form (route, timings) (FR18, FR21)	Admin on Add Train form	1. Leave route/timing fields blank 2. Submit	Route = ""	Error "Route and schedule details are required"	As expected	Pass
UT_24	UC8: Admin – Manage Trains, Routes & Schedules	Validate admin-only access control on train management pages	Logged in as non-admin user	1. Attempt to access train management URL directly	Role = "Passenger"	Access denied / redirected to home page	As expected	Pass
IT_22	UC8: Admin – Manage Trains, Routes & Schedules	Verify adding a train creates a record searchable by users (FR18)	Admin adds a new train	1. Add a new train 2. Search for it as a user	Train No. = "12999"	New train appears in user-facing search results	As expected	Pass
IT_23	UC8: Admin – Manage Trains, Routes & Schedules	Verify modifying train details updates the record used in search results (FR19)	Admin updates a train's timing	1. Update departure time 2. Search the same train	Train No. = "12658", New time = "14:30"	Updated timing shown to users in search results	As expected	Pass
IT_24	UC8: Admin – Manage Trains, Routes & Schedules	Verify removing a train removes it from user-facing search (FR20)	Admin deletes a train	1. Delete a train 2. Search for that train/route	Train No. = "99999" (test train)	Train no longer appears in search results	As expected	Pass
ST_22	UC8: Admin – Manage Trains, Routes & Schedules	End-to-end admin adds a new train with route and schedule successfully (FR18, FR21)	Logged in as admin	1. Navigate to Add Train 2. Fill route/schedule details 3. Submit	Train No. = "12999", Route = Bangalore-Pune	New train saved and searchable by users	As expected	Pass
ST_23	UC8: Admin – Manage Trains, Routes & Schedules	End-to-end admin modifies an existing train's schedule (FR19, FR21)	Logged in as admin	1. Open existing train 2. Update schedule 3. Save	Train No. = "12658"	Schedule change saved and reflected in search	As expected	Pass
ST_24	UC8: Admin – Manage Trains, Routes & Schedules	End-to-end non-admin user blocked from train management pages	Logged in as passenger	1. Attempt to navigate to admin train management page	Role = "Passenger"	Access blocked with "Unauthorized" message	As expected	Pass

UC9: Admin – View Bookings & Manage Seat Availability

Covers: FR22, FR23, FR24

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_25	UC9: Admin – View Bookings & Manage Seat Availability	Validate admin booking view filters correctly by train/date (FR22)	Bookings exist across trains/dates	1. Apply train/date filter 2. Check results	Train = "12658", Date = "18-09-2026"	Only matching bookings displayed	As expected	Pass
UT_26	UC9: Admin – View Bookings & Manage Seat Availability	Validate seat availability update function accepts only valid values (FR23)	Admin on seat management page	1. Enter a negative seat count 2. Submit	Seats = -5	Error "Enter a valid seat count"	As expected	Pass
UT_27	UC9: Admin – View Bookings & Manage Seat Availability	Validate booking/cancellation records retain all required fields (FR24)	Record created after a booking/cancellation	1. Inspect a stored record	Booking/Cancellation record	Record contains PNR, passenger, train, status & date fields	As expected	Pass
IT_25	UC9: Admin – View Bookings & Manage Seat Availability	Verify admin can view all passenger bookings for a selected train (FR22)	Bookings exist for a train	1. Open admin bookings view 2. Select a train	Train = "12658"	All passenger bookings for that train listed	As expected	Pass
IT_26	UC9: Admin – View Bookings & Manage Seat Availability	Verify admin seat availability update reflects on the user search page (FR23)	Admin updates seat count	1. Update available seats for a class 2. Check user search page	Train="12658", Class="SL", Seats=40	Updated seat count visible to users immediately	As expected	Pass
IT_27	UC9: Admin – View Bookings & Manage Seat Availability	Verify booking and cancellation records are persisted for audit (FR24)	Booking then cancellation performed	1. Book a ticket 2. Cancel it 3. Check records table	PNR = "1234567890"	Both booking and cancellation records retained in DB	As expected	Pass
ST_25	UC9: Admin – View Bookings & Manage Seat Availability	End-to-end admin views passenger bookings for a specific train (FR22)	Logged in as admin	1. Navigate to admin bookings view 2. Select a train	Train = "12658"	List of passenger bookings for that train displayed	As expected	Pass
ST_26	UC9: Admin – View Bookings & Manage Seat Availability	End-to-end admin manually adjusts seat availability for a train (FR23)	Logged in as admin	1. Open seat management 2. Adjust seat count 3. Save	Train="12658", Class="3A", Seats=20	Seat availability updated and reflected to users	As expected	Pass
ST_27	UC9: Admin – View Bookings & Manage Seat Availability	End-to-end verification that cancelled and confirmed records are both retained (FR24)	Mix of confirmed and cancelled bookings exist	1. Open admin records view 2. Check both statuses present	Multiple PNRs, mixed statuses	Both confirmed and cancelled records visible and intact	As expected	Pass